B.Tech. Degree VIII Semester Examination, April 2009

ME 802 MACHINE DESIGN II

(2002 Scheme)

Time: 3 Hours Maximum Marks: 100

I a) Explain uniform wear and uniform pressure assumption in Clutches? (5)

b) Sketch and explain working of internal expansion elements in Brakes? (5)

c) Two shafts whose centers are 100 cm apart are connected by a V-belt drive. The driving pulley is supplied with 125 H.P. and has an effective diameter of 30 cm. It runs at 1000 rpm. While the driven pulley runs at 375 rpm. The angle of grove on the pulleys is 40^0 Permissible Tension in 4 sq. cm. Cross-sectional area belt is 21 kgf/cm^2. The material of the belt weights 1.11 gram per cubic centimeter. The driven pulley is over hung, the distance of the center from the nearest bearing being 20 cm. The co-efficient of friction between belt & pulley rim is 0.28

Estimate
1. The number of belts required
2. Diameter of driven pulley shaft; if permissible shear stress is 420 kgf/cm^2 (15)

OR

II a) What are the advantages of V belt over a flat belt? (5)

b) Sketch & Describe a band type brake? (5)

c) Determine the maximum, minimum and average pressure in a plate clutch when the axial force is 4 kN. The inside radius of the contact surface is 50 mm and the outside radius is 100 mm. Assume uniform wear. (15)

III a) Sketch a gear profile (Spur) and mark all the terms used to represent different portions. (5)

b) Define :
1. Module
2. Diametral pitch
3. Circular pitch
4. Pressure angle (5)

c) A motor shaft rotating at 1500 rpm has to transmit 15 KW to a low speed shaft with a speed reduction of 3 : 1. The teeths are 14 ½ involute with 25 teeth on the pinion. Both the pinion and the gear are made of steel with a maximum safe stress of 200 N/mm^2. A safe stress of 40 N/mm^2 may be taken for the shaft on which the gear is mounted and for the key. Design a spur gear drive to suit the above condition. Assume the starting torque to be 25% higher than the running torque. (15)

OR

IV a) What are the merits and demerits of helical gears? (5)

b) What is interference in gears? How can you overcome it? (5)

c) A worm drive transmits 15 KW at 2000 rpm to a machine carriage at 75 rpm. The worm is Triple threaded and has 65 mm pitch diameter. The worm gear has 90 teeth of 6 mm module. The tooth form is to be 20^0 full depths in volute. The co-efficient of friction between mating teeth may be taken as 0.10. Calculate
1. Tangential force acting on the worm
2. Axial thrust and separating force on the worm and
3. Efficiency of worm drive. (15)

V a) With sketch explain nomenclature of a ball bearing? (5)

b) What are rolling contact bearings? (5)

c) A journal bearing 60 mm in diameter and 90 mm long runs at 450 rpm. The oil used for hydrodynamic lubrication has absolute viscosity of 0.06 kg/m.s. If the diameteral clearance is 0.1 mm. Find the safe load on the bearing? (15)

OR

(Turn over)

VI	a)	What are the principle consideration while selecting a bearing?	(5)
	b)	Write Notes: 1. Static and Dynamic Capacity 2. Materials for bearing	(5)
	c)	The thrust of propeller shaft is absorbed by 6 collars. The rubbing surface of these collars has outer diameter 300 mm and inner diameter 200 mm. If the shaft runs at 120 rpm, the bearing pressure amounts to 0.4 N/mm^2. The co-efficient of fraction may be taken as 0.05. Assuming that the pressure is uniformly distributed, determine the power absorbed by the collars.	(15)
VII	a)	What are the general design recommendations for rolled sections?	(10)
	b)	What are the main design considerations for forgings? Give details with sketches.	(15)

OR

VIII	a)	Write Notes on : 1. Tolerance 2. Surface finish	(10)
	b)	Discuss "Modification of Design" for manufacturing easiness in various manufacturing processors?	(15)

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